MILITARY SPECIFICATION SHEET

CABLE ASSEMBLY, RADIO FREQUENCY (AIRCRAFT AUDIO), CABLE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of specification MIL-C-22442.

APPLICABLE DOCUMENTS:

MIL-C-572	Cords, Yarns and Monofilaments Organic Synthetic Fiber
MIL-I-3930	Insulating and Jacketing Compounds, Electrical (for Cables, Cords and Wires)
MIL-C-11997	Cord Assembly, Electrical CX-2151( )/U (Retractile)
MIL-C-13273	Cord, Electrical (Retractile 2, 3 and 4 Conductor WD-9/U, WT-2/U and WF-4/U)
MIL-STD-104	Limit for Electrical Insulation Color

REQUIREMENTS:

1. All performance requirements and construction requirements except as modified by this specification sheet shall be in accordance with MIL-C-13273.

2. Conductors. Each conductor shall be equivalent to 404 circular mils (No. 24 AWG) and shall consist of 41 strands of No. 40 AWG tinned soft cadmium copper alloy, having a 1 percent nominal content of cadmium. It shall be tight bunch stranded with a lay not exceeding 0.75 inch. Prior to stranding, the elongation of individual tinned soft cadmium copper alloy strands shall be not less than 10 percent and the tensile strength shall not exceed 55,000 pounds per square inch. Subsequent to stranding and prior to insulating, the elongation of individual strands shall not be less than 5 percent and the tensile strength shall not exceed 58,000 pounds per square inch. Each conductor in the finished cord shall have a nominal DC resistance of 39.0 ohms per 1000 feet at or corrected to a temperature of 20°C.

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3. Insulation. The insulation for the cord shall be Type IR (or Type IH) rubber compound conforming to the requirements of MIL-I-3930 and capable of meeting the requirements specified herein.

4. Dimensions of insulated conductors. The diameter of the insulated stranded conductors shall be 0.057 inch nominal and shall not exceed 0.063 inch. The minimum wall thickness shall not be less than 0.009 inch.

5. Colors. Colors shall conform to MIL-STD-104. The color of the insulation on the conductors shall be black, blue, red, green, yellow and orange.

6. Separators. Separators shall consist of a serving, a knitted barrier, or a close braid of synthetic yarn conforming to Type P, AR or SAR of MIL-C-572 or of soft cotton. If, however, the insulating compound applied is free-stripping, the separator may be omitted.

7. Purity of yarns. The purity of yarns used as a separator under insulation shall be in accordance with MIL-C-11997.

8. Staycord. Staycords shall consist of synthetic textile yarn conforming to Type P, AR or SAR of MIL-C-572. The cord shall have a minimum breaking strength of 25 pounds and shall not abrade or otherwise damage the insulation when the cord is bent or stretched.

9. Binder. A binder, if used, shall consist of a suitable number of ends of synthetic textile yarns conforming to Type P, AR or SAR of MIL-C-572, or of soft cotton.

10. Shield. The shield shall consist of 32 nominal strands of No. 36 AWG tinned soft cadmium copper alloy having a 1 percent nominal content of cadmium and served over an insulated conductor in an even helix (laid adjacent to each other) with a nominal 90% coverage. A served shield shall be applied over each black, blue twisted pair and orange, yellow twisted pair.

11. Outer jacket. The cabled conductors shall be jacketed with a type JS-L(SBR), JH(EPDM), JN or JNL compound per MIL-I-3930 or a SBR natural rubber blend compound conforming to the performance requirements of MIL-C-13273. The amount of natural rubber in the jacket compound shall not exceed 35 percent by weight of the total SBR and natural rubber in the compound. The amount of sunproofing wax included in either compound shall be in accordance with MIL-I-3930.

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12. Jacketed cord. After vulcanization of the jacket the average overall diameter for the 6 conductor cord shall be .300 +.010 -.015 inch. The average diameter shall be taken as a mean of the maximum and minimum diameter of 10 measurements at various points along the cord. Diameters of retractile cords shall be taken along the straight portions of the cord at both ends. The jacket thickness at any cross section of the cord shall be .010 minimum. The jacket shall be free-stripping from the insulation. Adapter separator may be used under the jacket.

13. Construction. The cable shall be constructed of 6 insulated conductors, orange and yellow twisted together with a shield, black and blue twisted together with a shield, a metal braid, and a red and a green. The 6 conductors shall be twisted around a stay cord. Filler may be added if necessary to make a circular cross section. The lay of the conductors shall be 1.75 inch maximum. A suitable binder or separator may be applied over the cabled conductors.

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-22442/34(AS)	CABLE ASSEMBLY, RADIO FREQUENCY (AIRCRAFT AUDIO), CABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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